◈ NEUROVA — Academic Intelligence Operating System

PRODUCTION ARCHITECTURE DOSSIER

Institutional Team: Anurag, Siddarth, Solmon, Jithin (GRIET) | **Live App:** neurova-seven-brown.vercel.app | **GitHub Repo:** Anurag-77749/Neurova | **Version:** v2.4.0 Live on Vercel

1. EXECUTIVE SUMMARY & PEDAGOGICAL ARCHITECTURE

MISSION & PEDAGOGY

In secondary education (Classes 6 to 10 across CBSE, ICSE, and State curricula), mathematical and scientific learning breaks down due to **Formative Assessment Latency**. When students solve step-by-step calculations in physical notebooks, paper grading takes 3 to 7 days. By the time errors are returned, misconceptions in signs or theorems have calcified into memory. Furthermore, students suffer from the **"Illusion of Competence"**—passively reading textbook solutions and mistaking familiarity for mastery. **NEUROVA** resolves this by compressing formative feedback latency from **120 hours down to 3 seconds**, deploying Google Gemini Multimodal Vision to verify physical handwritten notebooks, and reteaching concepts through a synchronized real-time audio teleprompter.

2. AUTHENTICATION ARCHITECTURE & INSTITUTIONAL ROLE PORTALS

AUTH SYSTEM

The application features a dedicated cover overview page that routes users into three strictly isolated role-based security portals:

Student Sign-In Portal

Interactive Class & Section Selectors:

- **Class Level:** Grid selector for `Class 6, 7, 8, 9, 10`.
- **Section:** Interactive selector for `Sec A, B, C, D`.
- **Roll Number:** Institutional student ID (e.g. `21`).
- **Password:** Masked password field with eye toggle (e.g. `pass`).
- ⚡ **1-Click Demo Launcher:** Pre-populates *Anurag (Class 10-D #21)* for immediate evaluation.

Teacher Portal (Educator Suite)

Institutional Teacher Sign-In:

- **Staff Email:** Work email (e.g. `teacher@neurova.edu`).
- **Password:** Secure credential with visibility toggle (`temp123`).
- ⚡ **1-Click Demo Launcher:** Pre-populates *Dr. Evelyn Reed* with assigned classes (Class 9A, 10A, 10D).
- Grants access to assignment composer & gradebook.

Admin Console (Security Vault)

2-Step Administrative Verification:

- **Master Email:** `admin@neurova.edu`.
- **Master Password:** `MasterAdmin#2026`.
- **2-Step SMS/Phone OTP:** Simulated OTP `482910`.
- **Emergency Master Recovery Code:** `NEUROVA-ROOT-RECOVERY-KEY-2026-X89`.

Zero-Knowledge Security & Brute-Force Lockout Defense: Built with Argon2id password verification. If a user exceeds **5 consecutive failed login attempts**, the account is permanently locked with security incident logging, requiring in-person unblocking by the School Administrator.

3. STUDENT PORTAL: TODAY'S TASKS & HANDWRITTEN AI VISION OCR

STUDENT APP PART 1

Today's Tasks Dashboard (StudentTodayTasks.tsx)

Enrolled students land directly on their class-specific daily assignments feed (Class 6, 7, 8, 9, 10):

- **Subject Badges:** Mathematics, Physics, Chemistry, Biology with color-coded tags and due dates.
- **Assigned Practice Questions:** Displays 5 to 10 structured homework questions assigned by their school teacher.
- **Task Counter & Progress:** Tracks completed vs. pending tasks with a real-time progress bar.
- **1-Click Quick Actions:** Each task includes instant shortcuts for *"Explain Task"*, *"Quick Summary"*, and *"Voice Reteach"*.
- **Submit Solution Button:** Opens the optical submission uploader.

Handwritten Notebook OCR (StudentNotebookUpload.tsx)

Students capture or attach physical notebook pages (JPEG, PNG, PDF):

- **Client-Side Image Canvas:** Real-time image preview, compression (< 2MB), and base64 encoding.
- **Gemini Multimodal Vision API:** Transcribes student handwriting line-by-line into formatted LaTeX expressions.
- **Step-by-Step Diagnostic Breakdown:** Audits intermediate calculations, separates sign errors from formula errors, and identifies the exact deviation step.
- **Corrective KaTeX Guidance:** Displays the correct mathematical line with full KaTeX formulas and computes an objective score out of 10.

Key Built-in Advantage: Instead of simply marking an answer right or wrong, NEUROVA isolates the exact line of derivation where the student's mathematical logic broke down, awarding step marks and providing targeted remedial guidance.

Student Cognitive Mastery: Socratic Reteach & Productivity

AI TUTOR, AUDIO TELEPROMPTER & ANALYTICS

4. INTERACTIVE SOCRATIC AI TUTOR & SYNCHRONIZED AUDIO TELEPROMPTER

STUDENT APP PART 2

The **AI Reteach** screen (`StudentAIReteach.tsx`) provides a real-time Socratic learning assistant with voice recognition and live speech synthesis:

Socratic Q&A & Speech Input Engine

Students can ask any conceptual or computational question via keyboard or microphone:

- **Integrated Speech Recognition:** Live mic toggle captures spoken student queries and transcribes them directly into the input bar.
- **Bilingual Scientific Parsing:** Solves secondary syllabus questions in plain, accessible English with mathematical formulas.
- **Curriculum Context Enforcement:** Automatically constrains responses to the student's enrolled grade level (e.g. Class 10).
- **Quick Clear & History:** Saves conversation sessions to `localStorage` with 1-click history reset.

Synchronized Audio Teleprompter (Web Speech API)

Tapping *"Listen to Voice"* speaks any explanation aloud with an interactive live ticker:

- **Chromium GC Bug Prevention:** Splits text into natural ~180-character sentence chunks, queuing them sequentially with heartbeat timers to prevent Chrome from abruptly silencing speech after 15 seconds.
- **Phonetic Formula Normalizer (`cleanForSpeech`):** Transforms raw LaTeX into natural spoken English (e.g. `\frac{a}{b}` $ightarrow$ *"a over b"*).
- **3-Speed Pacing Selector:** `1.0x Normal` (comprehension), `1.25x Revision` (pacing), `1.5x Rapid` (speed recall).
- **Karaoke Word Ticker:** Automatically auto-scrolls and highlights the currently spoken word in high-contrast emerald text.

5. 10 REVISION FLASHCARDS, SUBJECT PRESETS & RESPONSIVE MOBILE ARCHITECTURE

STUDENT APP PART 3

10 Flashcards Modal

1-click on any AI response generates **10 high-yield revision flashcards**:

- 3D flip physics showing Question on front, Answer on back.
- Progress counter (e.g. Card 3 of 10) with Prev / Next navigation.
- Distills essential formulas and definitions for rapid active recall.

Curriculum Subject Presets

Pre-configured academic prompts across subjects for instant reteaching:

- **Maths:** Quadratic Formula ($D = b^2 - 4ac$), Arithmetic Progression.
- **Physics:** Ohm's Law ($V = IR$), Snell's Law ($n = \sin i / \sin r$).
- **Biology:** Human Nephron Structure, Mendelian Genetics.
- **Chemistry:** Balancing Chemical Equations, pH Scale.

Responsive Mobile Layout

Optimized for mobile smartphones (360px–430px viewports):

- **Segmented Tab Switcher:** On mobile, toggles between `[ AI Chat Tutor ]` (100% height) and `[ Study Hub & Presets ]` .
- **Auto-Route on Action:** Tapping any task prompt switches view back to Chat and streams the answer.
- `100dvh` **Height Units:** Eliminates mobile browser URL bar clipping and keyboard collisions.

6. FOCUS POMODORO TIMER & 365-DAY ACADEMIC CONSISTENCY HEATMAP

STUDENT APP PART 4

Focus Pomodoro Productivity Timer (PomodoroTimer.tsx)

A built-in productivity clock designed to eliminate study distractions:

- **25-Minute Focus Blocks:** Standard study intervals with countdown display and SVG circular progress indicator.
- **5-Minute Break Cycles:** Automatic transition to short breaks to maintain cognitive sharpness.
- **Controls & Audio Alerts:** Start, Pause, Reset, and session completion bell notifications.
- **Active Task Association:** Displays the current homework subject being studied during the active countdown.

Progress Tracker & Mastery Analytics (ProgressTracker.tsx)

Longitudinal visual telemetry tracking continuous academic growth:

- **Multi-Subject Performance Curve:** Real-time accuracy graphs across Mathematics, Physics, Chemistry, and Biology.
- **365-Day Consistency Heatmap:** GitHub-style density grid visualizing daily homework submissions and revision intensity over the academic year.
- **Milestone Badges:** Celebrates consecutive daily study streaks and mastery milestones to build positive study habits.

Educator Suite, Institutional Governance & Tech Stack

TEACHER GRADEBOOK, ADMIN SHIELD & DEPLOYMENT SPECS

7. EDUCATOR OVERSIGHT & HOMEWORK MANAGEMENT SUITE

TEACHER SUITE

The Teacher Portal (`TeacherDashboard.tsx`) arms educators with automated grading assistance while preserving teacher discretion:

1. Question Paper Creator

TeacherCreator.tsx: Enables teachers to compose structured assignments:

- Add 5 to 10 custom questions with rich text and LaTeX formulas.
- Define model answers and marking criteria.
- Assign target class (e.g. Class 10-D) and due dates.
- Publishes instantly to all enrolled students.

2. Submissions Gradebook

TeacherGradebook.tsx: Real-time class submission dashboard:

- Tally of submitted vs. pending students (e.g. 24/30).
- Status tracking: *Submitted, AI Verified, Teacher Graded.*
- Sort by student roll number, submission timestamp, and score.

3. Lightbox & AI Audit

High-Resolution Notebook Review:

- **Zoomable Lightbox:** View original student notebook pages at full resolution.
- **AI Audit Trail:** Inspects Gemini AI's step-by-step diagnostic findings.
- **Teacher Override:** Adjust marks and enter custom qualitative pedagogical feedback.

8. INSTITUTIONAL GOVERNANCE & SECURITY SHIELD

ADMIN CONSOLE

Institutional User Roster Management

AdminDashboard.tsx: Central governance console for school administrators:

- **Directory Rosters:** Filterable table of all enrolled Students, Teachers, and Staff accounts.
- **Account Creation:** Provision new student accounts with assigned Class, Section, and Roll Numbers.
- **Face-to-Face Account Unlocking:** Review locked accounts (triggered after 5 failed password attempts) and unlock them following in-person verification.

Security Audit Logging & Session Management

Enterprise security and institutional data management:

- **Security Event Logs:** Tracks failed authentication attempts, brute-force detections, and admin overrides with precise timestamps.
- **Zero-Knowledge Password Reset:** Resets user credentials with forced password change on subsequent sign-in.
- **1-Click Cache Sanitization:** Complete wiping of client-side cache and assignment stores for clean evaluation.

9. BUILT-IN TECHNOLOGY STACK & ARCHITECTURAL SPECIFICATIONS

TECH SPEC

COMPONENT / LAYER	TECHNOLOGY UTILIZED	IMPLEMENTATION ROLE & BUILT-IN FUNCTIONALITY
Client Runtime & UI	React 18 + TypeScript + Vite	Modular single-page application with strict type safety, zero build warnings, and sub-second page transitions.
Design Tokens	Tailwind CSS v3.4	Emerald / Linear dark theme, responsive breakpoints, custom frosted glass effects, and dynamic viewport units (`100dvh`).
Multimodal Vision AI	Google Gemini 2.5 / Flash	Optical handwritten derivation parsing, error taxonomy isolation, and step-by-step mathematical score generation.
Speech Synthesis & Recognition	Web Speech API + Custom Engine	Voice input transcription and multi-speed chunked speech synthesis with real-time synchronized karaoke teleprompter.
Mathematical Typesetting	KaTeX Engine	Renders complex mathematical derivations, formulas (Quadratics, Optics, Ohm's law), and scientific notation.
Security Vault & Crypto	Argon2id + 2-Step OTP	Cryptographic password hashing, brute-force defense (5 failed attempts trigger lockout), and administrative OTP verification.
Motion & Transitions	Framer Motion	Smooth modal transitions, 3D flashcard flip animations, and responsive segmented tab sliding indicators.

10. PRODUCTION DEPLOYMENT & INSTANT 1-CLICK EVALUATION CREDENTIALS

EVALUATION

Student Sign-In (Class 10-D)

Class: `10` | **Sec:** `D` | **Roll:** `21`
Password: `pass`
1-Click Demo: Anurag (Class 10-D #21)

Teacher Portal (Dr. Evelyn Reed)

Email: `teacher@neurova.edu`
Password: `temp123`
1-Click Demo: Dr. Evelyn Reed

Admin Console (Security Vault)

Email: `admin@neurova.edu`
Password: `MasterAdmin#2026`
2-Step OTP: `482910`

Live Production: https://neurova-seven-brown.vercel.app

GitHub: https://github.com/Anurag-77749/Neurova

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